ELCO DRIL-FLEX SELF DRILLING FASTENERS

Cat #	Description	Pt.	Load-Bearing Area	Finish	Ctn Qty	Ctn Wt

#10 Diameter, 5/16" Hex Washer Head

EAF430	10-16 x 3/4"	#3	.380"	Stalgard SUB	6,000	48
EAF460	10-16 x 1-1/2"	#3	1.25"	Stalgard SUB	2,500	30
EAF470	10-16 x 2"	#3	1.50"	Stalgard SUB	2,000	28
EAF480	10-16 x 2-1/2"	#3	2.25"	Stalgard SUB	1,500	25

#10 Diameter, #2 Phillips Pan Head

EDX445	10-16 x 3/4"	#2	.380"	Stalgard	6,000	36

#10 Diameter, #2 Phillips Wafer Head

EBL530	10-24 x 1-1/4"	#3	.750"	Stalgard	5,000	44

#12 Diameter, 5/16" Hex Washer Head

EAF621*	12-14 x 7/8"	#3	.380"	Stalgard SUB	5,000	54
EAF641	12-14 x 1"	#3	.500"	Stalgard SUB	4,000	44
EAF661	12-14 x 1-1/4	#3	.750"	Stalgard SUB	2,500	36
EAF681	12-14 x 1-1/2"	#3	1.00"	Stalgard SUB	2,500	38
EAF690	12-14 x 2"	#3	1.50"	Stalgard SUB	2,000	37
EAF715	12-14 x 3"	#2	2.35"	Stalgard SUB	1,000	27
EAF755	12-24 x 1-3/4"	#5	.800"	Stalgard SUB	2,500	45

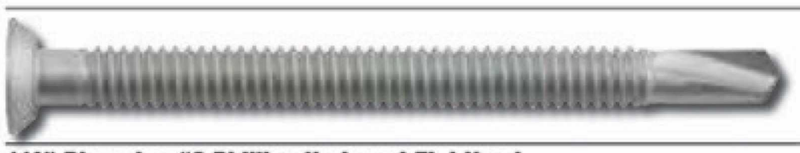
#12 Diameter, #3 Phillips Undercut Flat Head

EBL215	12-14 x 1"	#3	.500"	Stalgard	4,000	36
EBL220	12-14 x 1-1/4"	#3	.750	Stalgard SUB	3000	29
EBL223	12-14 x 1-1/2	#3	1.00"	Stalgard	2,500	26

1/4" Diameter, 3/8" Hex Washer Head

EAF816	1/4-14 x 1"	#3	.450"	Stalgard SUB	3,000	54
EAF841	1/4-14 x 1-1/2"	#3	.950"	Stalgard SUB	2,000	45
EAF876	1/4-20 x 1-1/2"	#4	.830"	Stalgard SUB	2,000	48
EAF846	1/4-14 x 2"	#3	1.45"	Stalgard SUB	1,500	41

EAF886	1/4-20 x 2"	#4	1.33"	Stalgard SUB	1,500	45
EAF865	1/4-20 x 1-1/8"	#4	.500"	Stalgard SUB	2,500	51
EAF888	1/4-20 x 1-3/4"	#5	.800"	Stalgard SUB	1,000	27
EAF890	1/4-20 x 2-1/2"	#4	1.83"	Stalgard SUB	1,000	45
EAF900	1/4-20 x 3-3/8"	#4	2.70"	Stalgard SUB	500	22
EAF910	1/4-20 x 4"	#4	3.50"	Stalgard SUB	500	23



1/4" Diameter, #3 Phillips Undercut Flat Head

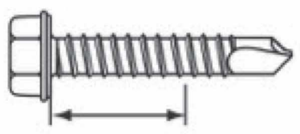
EBL330	1/4-20 x 3"	#4	2.50"	Stalgard	500	20
EBL340	1/4-20 x 4"	#4	3.50"	Stalgard	500	23

5/16" Diameter, 3/8" Hex Washer Head

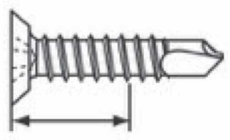
EAF940	5/16-18 x 1-1/2"	#3	.800"	Stalgard SUB	1,000	37
EAF960	5/16-24 x 1-1/2"	#4	.800"	Stalgard SUB	1,000	40
EAF970	5/16-24 x 2"	#4	1.25"	Stalgard SUB	1,000	43

* for aluminum applications only

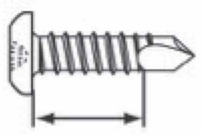
Load Bearing Area



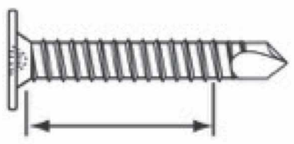
Hex Washer Head



Undercut Head



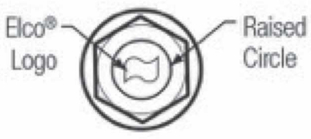
Pan Head



Wafer Head

Identification

(hex washer head shown)



Comparison to Stainless Steel Screws

Galvanic Series

Anodic End	
Metal / Alloy	**EMF(v)**
Magnesium	-1.60
Zinc	-1.10
Alum (5000, 6000, 7000)	-.75
Iron, Low Alloy Steels	-.70
Alum (2000)	-.60
Lead	-.55
18% Chromium Steel	-.35
Naval Brass	-.30
Brass, Bronze	-.25
Austenitic Stainless (300 Series)	-.20
Nickel	-.15
Silver	0
Gold	+15
Cathodic End	

300 series stainless steel fasteners provide high resistance to hydrogen embrittlement failures. However, stainless steel is galvanically incompatible with aluminum or steel panels. In this case, stainless steel fasteners trigger a sacrificial action, which can lead to degradation of the panel and loosening of the fastener.

A dual-hardening process allows Dril-Flex® fasteners to provide high strength and resistance to hydrogen embrittlement failures. Their Stalgard® finish provides corrosion resistance several times greater than other commonly-used finishes. If an environment is corrosive enough to significantly affect the Stalgard finish, the potential for significant degradation of the aluminum/stainless steel assembly would also exist.

Embrittlement Tests

Embrittlement testing of Dril-Flex screws was performed in accordance to ASTM F1624-06. Fastener lots were tested to determine their Threshold Stress Limits for both Internal Hydrogen Embrittlement and Environmental Hydrogen Embrittlement. Threshold Stress Limit is the stress level below which no time-dependent cracking will occur. Above this level, subcritical cracking that leads to time-delayed fracture or embrittlement may occur if the fastener is exposed to a hydrogen environment.

Embrittlement Test Results

- Dril-Flex fasteners have a hardness range of HRC 28 – 34, which is roughly equivalent to a SAE Grade 5 fastener (HRC 25 – 34).
- Dril-Flex fasteners showed resistance to the effect of hydrogen-assisted cracking when loaded to 75% of their tensile strength. This is within accepted industry guidelines for in-service loading conditions.
- Dril-Flex fasteners showed no degradation or failures in tensile strength below their ultimate tensile strength.